DRUMPANEL · STEM CHECK

DrumPanel STEM check

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First STEM check (kit 1.8 upgrade of a TRL 3 repo); 28 claims for option A from DMP-CAL-001 and 8 first-order claims for option B; independent review recorded; five failed details fixed under Amish's standing instruction of 2026-10-23 (DMP-DEC-001, F1 to F5)

DrumPanel STEM check

Verdict: every claim the bench set makes sits within known science once five details found by the independent review are fixed, and nothing fails, so `stem_check` is **pass**. The review refuted the claim that one bending roll reading holds for a batch of drums and found four more details that failed: one drain stand could not purge drums fast enough, a methane-calibrated LEL sensor can under-read heavy vapour, the 8 mm guard slots sat on the edge of ISO 13857 Table 4, and fire-resistant hydraulic fluids do not settle out of purge water. Each was fixed in the design under Amish's standing instruction of 2026-10-23 (DMP-DEC-001, fixes F1 to F5) and its row closed. Of the 28 claims for option A, the current design, 23 are within known science and 5 are plausible and closed by the calculation note DMP-CAL-001 on paper (the two hardest hand forces, 136 N and 131 N against 150 N, the force limit itself, the single driven disc's traction and the cut zone temperature); each rests on a screening value that the TRL 4 trial measures. The 8 claims for option B, the lever shear and stretch leveller, are first-order screens and within known science. Method: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py`, which takes its option A numbers from `docs/04-calcs/sizing.py`.

> **Safety:** The STEM check shows that the physics and chemistry hold on paper. It does not show that a given drum is free of vapour: the vapour check on every drum, every day, with a detector calibrated for heavy hydrocarbon vapour, is the safeguard (S4), and drums that held fuel, solvent, chemicals, fire-resistant hydraulic fluids or unknown contents are never processed.

Claims checked

Table 1. Claims, governing science and verdicts (generated from `docs/04-calcs/stem_results.csv`; S rows are option A, B rows option B).

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	Each drum gives three flat panels with about 1.26 m2 of sheet	Mathematics	Units and dimensional analysis	Panels = mid-wall circumference x body length left after the chime rings and hoop strips are cut out (0.70 x 1.80 m); it must be less than the whole body between the chimes; the independent review found the README's three sheets of 460 x 1,830 mm per drum (2.5 m2) cannot come from one drum, and the README is corrected	1.26 m2 <= 1.53 m2	Within known science	DMP-CAL-001
S2	A drum full of water weighs about 240 kg, so it is filled and drained on its stand and never lifted	Physics	Mechanics and statics	Mass = water volume (pi r2 L) x 1,000 kg/m3 + empty drum	claimed 240.146 kg, estimate 240 kg (ratio 1.00)	Within known science	DMP-CAL-001
S3	A full drum drains through its 2 in bung in about 2.3 min	Physics	Fluid mechanics	Torricelli outflow Q = Cd a sqrt(2 g h), Cd 0.6, integrated over the falling level of a horizontal cylinder to the bung centre (the CAL adds the slow tail to the lip); the top bung is open as the vent	claimed 2.3 min, estimate 1.77 min (ratio 1.30)	Within known science	DMP-CAL-001
S4	A water fill and drain, then a pumped LEL reading below 5 % LEL at both	Chemistry	Physical chemistry (flammability)	29 CFR 1926.352(i): containers that held flammable substances are filled with water or cleaned,	Pass level 5 % LEL is half the 10 % hot-work limit; the oils accepted have flash points above 60 C (GHS category 3 upper	Within known science	DMP-DEC-001 fix F3

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	bungs, makes a drum that held a non-volatile oil safe to cut cold			ventilated and tested before cutting; 29 CFR 1910.252(a)(3); hot work barred at or above 10 % of the LEL (29 CFR 1915.14(b)); AWS F4.1:2017 water cleaning	bound), so little vapour forms at workshop temperatures; the reading, not the purge, is the safeguard. The independent review found that a catalytic sensor calibrated on methane under-reads heavy hydrocarbon vapour (relative response about 0.3 to 0.5), so a true 10 to 15 % LEL could read 5 %: fixed by specifying the LEL channel calibrated on pentane or set with the maker's correction factor (BOM line 4, PR-002)		
S5	The end cutter takes a 1.5 mm head out with a crank force of at most 27 N	Physics	Strength of materials	Wheel shear F = tau t2 / (2 tan theta) against the chime wall, feed resistance F tan(theta / 2), 30 % roller losses, 150 mm crank	27 N <= 150 N	Within known science	DMP-CAL-001
S6	Two ring cuts at once on a 1.5 mm wall need at most 136 N at the shear head crank	Physics	Mechanics and statics	Rotary shear F = tau t2 / (2 tan theta), 1.0 mm overlap; edge spreading round the web 100 N at 1.2 mm (estimate, cubic in t); 20 % bearing losses; 2.33 to 1 drive, chains 95 % each;	136 N <= 150 N	Plausible, to confirm at TRL 3	DMP-CAL-001 [D15r]; the spreading force is to measure on a scrap drum at TRL 4

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				155 mm crank (CAL section 4)			
S7	The lever notching punch cuts the five-layer chime notch with at most 131 N on its 500 mm ratchet lever	Physics	Strength of materials	Inclined-edge shear per edge 0.5 tau t2 / tan(rake), rake 10 deg, five layers, 20 % stripping; Tr24 x 5 screw, friction 0.15 (CAL section 9); a flat punch would need about 177 kN, so the rake does the work	131 N <= 150 N	Plausible, to confirm at TRL 3	DMP-CAL-001 [K3]; the chime notch force is the least certain number, to confirm on a scrap chime at TRL 4
S8	The shear head frame and shafts carry the separating force at 1.5 mm with a large margin	Physics	Strength of materials	Web in bending and tension, 25 mm shaft in bending at 1,429 N (CAL section 4); S275 yield 275 MPa, used at 0.6 of yield	17.7 MPa <= 165 MPa	Within known science	DMP-CAL-001
S9	The bending roll setting that leaves a panel flat (9.3 to 19.3 mm above the lower roll) lies inside the 60 mm slot travel	Physics	Strength of materials	Elastic and perfectly plastic reverse bending: x - M(x)/Me = k0 / ke, Me = sy b t2 / 6, springback to flat; yield 200 to 300 MPa, wall 0.8 to 1.2 mm (CAL section 5.2)	19.3 mm <= 60 mm	Within known science	DMP-CAL-001
S10	With the first panel of every drum checked and the dial trimmed, the slip roll leaves panels flat within 10 mm over 1 m	Engineering and technology	Manufacturing tolerances	Setting allowance 0.29 mm from the derivative of the residual curvature (CAL section 5.3); a change of only 0.02 mm of wall or 5 MPa of yield uses it up, so a setting found once per batch (as first claimed) does not hold from drum to	0.025 mm <= 0.073 mm	Within known science	DMP-DEC-001 fix F1; DMP-CAL-001 sections 5.3 and 8

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				drum (independent review, recomputed in CAL section 5.3): fixed by a straightedge check of the first panel of every drum, a dial trim, and a second pass, turned over if over-bent, for a panel out of tolerance; the dial resolution must be at most a quarter of the allowance			
S11	Cutting the hoop strips out keeps the slip roll to plain sheet; rolled flat, a hoop crown would stretch about 8.5 %	Physics	Strength of materials	Crown strain = hoop height x change of curvature, against the 0.35 % surface strain of a plain panel in the slip roll: a strain more than ten times larger cannot be put in evenly by three rolls, so a rolled hoop would leave the panel buckled (DMP-DDR-002, P7); a design rationale: the check shows the crown strain is above ten times the panel strain	8.5 % >= 3.542 %	Within known science	DMP-CAL-001
S12	The geared pinch rolls grip three panels with enough force to drive them	Physics	Mechanics and statics	Pinch force needed = 1.5 x feed resistance / (2 x friction 0.15) at the worst steel (300 MPa, 1.2 mm); two M16	2000 N >= 1193 N	Within known science	DMP-CAL-001

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	through the bend			screws set 2 kN (CAL section 5.5)			
S13	The single driven shear disc pulls the head along the cut at 1.5 mm wall	Physics	Mechanics and statics	Traction 0.3 x separating force (the edge bites) against the feed resistance (CAL section 4); below 1 the operator pushes the trolley	1.17 of the resistance >= 1 of the resistance	Plausible, to confirm at TRL 3	DMP-CAL-001 [D15t]; friction 0.3 is a screening value, the fallback is a hand push on the trolley
S14	Every piece one person lifts from bench height is 25 kg or less; heavier pieces, and the shear head and ring head lifted to the rail, are two-person lifts	Biology	Ergonomics	ISO 11228-1 reference mass 25 kg, which holds only for ideal lifting; the independent review put the recommended limit for lifting the 23.7 kg shear head up to the rail beam at about 10 kg, so the build plan and PR-011 make both heads two-person lifts; masses from the station model (CAL section 10)	23.7 kg <= 25 kg	Within known science	DMP-CAL-001
S15	Every crank and lever force in normal use is at or below 150 N	Biology	Human physiology and biomechanics	Forces from S5 to S7 and the slip roll crank (20 N at 300 mm, CAL section 5.5); 150 N is the requirement PR-006, a conservative screen for a two-hand crank in a standing posture (EN 1005-3 assessment at TRL 4)	136 N <= 150 N	Plausible, to confirm at TRL 3	DMP-CAL-001; EN 1005-3 assessment at TRL 4
S16	Slots made 7 to 8 mm wide		Standards practice	ISO 13857:2019 Table 4: a slot	20 mm <= 45 mm		DMP-CAL-001;

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	and at least 45 mm from the slip roll nip keep fingers out	Engineering and technology		over 6 mm and up to 8 mm needs at least 20 mm to the danger point (CAL section 7); a slot made over 8 mm would need 80 mm, so the slots are toleranced 7 to 8 mm (fix F4, PR-010)		Within known science	DMP-DEC-001 fix F4
S17	Cold cutting stays cold: slow shearing of 1.5 mm wall does not heat the cut zone toward an oil's autoignition temperature	Physics	Thermodynamics	Plastic work tau t x t per metre plus feed work 365 J, into a 2 mm wide strip (c = 460 J/kg K) with no heat flow away; the independent review showed this is not a bound, since a narrower shear zone (0.2 to 0.9 mm) gives a higher local rise; the cut moves at about 4 m/min, so heat spreads into the sheet; lubricating oils autoignite above about 250 C (220 K above a 30 C workshop); the safeguard is the vapour reading below 5 % LEL (S4), and no grinder or torch is used	94 K <= 220 K	Plausible, to confirm at TRL 3	DMP-STM-001 S4 (the vapour reading is the safeguard); cut edge temperature to measure at TRL 4
S18	Hardened D2 discs and a hardened cutter wheel shear 0.8 to 1.5 mm low-carbon drum	Chemistry	Materials and polymer chemistry	Slitter knives for low-carbon sheet are commonly D2 tool steel hardened to about 58 to 62 HRC (cutting tool	Within practice; the discs bought are to the BOM line 10 specification (DMP-DEC-001,	Within known science	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	steel without chipping			supplier practice); the drum steel is about 250 MPa yield	to confirm when bought, item 1)		
S19	Purge water settles and its oil is skimmed off, so the water can be reused	Chemistry	Environmental and atmospheric chemistry	Mineral and vegetable oils are less dense than water (about 850 to 920 kg/m3) and separate by gravity; a biodegradable detergent emulsifies some oil, so the water settles overnight before it is reused; the independent review found that fire-resistant hydraulic fluids (phosphate esters, water glycols, polyalkylene glycols) sink or mix with water: they are excluded at acceptance (fix F5, PR-013)	Within known science; disposal of the skimmed oil follows local rules (DMP-REG-001)	Within known science	DMP-DEC-001 fix F5
S20	The process has no burning step, so it gives off none of the smoke of burning paint and oil residue	Biology	Toxicology	Open burning of coated and oily scrap releases smoke, soot and products of incomplete combustion; the bench set never heats the steel (PR-001)	Removed by design; the paint stays on the panels and is removed later by a separate cold step (DMP-DDR-001, D4)	Within known science	Open
S21	The bench set removes the chisel blows: its cuts and rolling are continuous,	Physics	Acoustics and vibration	Impact noise comes from blows on sheet steel; continuous rotary shearing and rolling at hand speed make	Within known science as a mechanism; sound levels are not claimed and are measured at TRL 4	Within known science	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	and the only impact step is setting the panel ends with a dead-blow mallet			no impacts (CAL section 5.4)			
S22	The eccentric bush sets the 1.0 mm disc overlap within its adjustment range	Engineering and technology	Manufacturing tolerances	A 3 mm eccentric moves the upper shaft over a 6 mm range; the overlap is set and locked (DMP-DDR-002, P4)	1 mm <= 6 mm	Within known science	DMP-CAL-001
S23	A cradle roller shaft carries the drum and a person leaning on it (1 kN) well below yield	Physics	Strength of materials	Simply supported 25 mm shaft, two roller loads (CAL section 6); bright steel taken at S275 yield 275 MPa, used at 0.6 of yield	72 MPa <= 165 MPa	Within known science	DMP-CAL-001
S24	Two people turn about 2.20 drums an hour into flat panels (PR-007 asks 2.5)	Mathematics	Orders of magnitude	Rate = 2 x 60 / hands-on minutes per drum (54.6 min, CAL section 8), with the per-drum flatness check of fix F1; task times are estimates; checks the arithmetic only, the timed trial measures the times	in 120 min, out 120 min	Within known science	Open
S25	Two drain stands purge drums faster than the cradle and slip roll use them	Mathematics	Orders of magnitude	Stand cycle 35.3 min a drum (load, fill 11 min, soak 10 min, drain, tip, check; CAL section 8); the independent review found one stand would limit the set to about 1.7 drums an hour: fixed by a second stand and	3.39 drums/h >= 2.2 drums/h	Within known science	DMP-DEC-001 fix F2

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				tray (fix F2, BOM lines 1 and 2)			
S26	The 60 mm bending roll bends little enough under its worst load that all three panels see the same setting	Physics	Strength of materials	Simply supported roll over 940 mm under 805 N spread along it; mid-span deflection against half the 0.29 mm setting allowance (CAL section 5.5)	0.068 mm <= 0.147 mm	Within known science	DMP-CAL-001
S27	The notching punch C-frame jaws stay well below yield	Physics	Strength of materials	Cantilever jaw root bending at the hoop station (CAL section 9); S355 yield 355 MPa used at 0.6 of yield	96 MPa <= 213 MPa	Within known science	DMP-CAL-001
S28	The rail beam keeps the shear head on a straight line with both heads on it	Physics	Strength of materials	Mid-span deflection of the 80 x 40 x 3 RHS beam under the trolley, both heads and drive (CAL section 4.1), against the 1 mm straightness of PR-004's rail	0.26 mm <= 1 mm	Within known science	DMP-CAL-001
B1	Option B: one person cuts 1.5 mm drum steel with the 1,050 mm shear lever	Physics	Mechanics and statics	Inclined blade shear 0.5 tau t2 / tan(rake), rake 8 deg: 2.31 kN; lever and link advantage 22.3, efficiency 0.85; lever throatless shears of this class are sold for 14 gauge (1.9 mm) mild steel (Beverly B-1)	122 N <= 150 N	Within known science	Open
B2	Option B: a 15 t hand-pumped cylinder stretches the widest,	Physics	Strength of materials	F = flow stress x area; 300 MPa yield x 1.1 for strain hardening	92.7 kN <= 147 kN	Within known science	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	thickest panel 2 % past yield			at 2 %, 234 x 1.2 mm			
B3	Option B: a 2 % stretch takes out the drum's curl	Physics	Strength of materials	Stretcher levelling: once the whole section yields in tension the bending stresses that hold the curl are wiped out; needs total strain above curl strain t / 2R (0.21 %) plus yield strain (0.15 %); levellers use 1 to 3 %	2 % >= 0.3594 %	Within known science	Open
B4	Option B: the hand pump handle needs at most 150 N at the stretch force	Biology	Human physiology and biomechanics	Pressure = force / plunger area (52 mm): 44 MPa; pump piston 8 mm, handle ratio 25, pump efficiency 0.8 (to confirm on the pump bought)	110 N <= 150 N	Within known science	Open
B5	Option B: the two UPN 100 bed channels carry the stretch force in compression	Physics	Strength of materials	Axial stress = F / (2 x 1,350 mm2), ties at 550 to 650 mm keep the channels short against buckling; S275 used at 0.6 of yield	34.3 MPa <= 165 MPa	Within known science	Open
B6	Option B: the wedge grips tighten as the panel pulls	Physics	Mechanics and statics	A wedge grip holds when the face friction on the panel (serrated, taken as 0.3) exceeds tan(wedge angle) plus the friction on its greased back (0.1); the first check (tan(angle) below 0.3, self-locking)	0.223 <= 0.3	Within known science	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				was the wrong criterion (independent review), so the wedge angle is 7 deg; the panel's net section at the serrations is to check if B is chosen			
B7	Option B: each panel loses its gripped ends	Mathematics	Units and dimensional analysis	Panel length after stretching = 1,800 - 2 x 60 mm grip + 2 % (PR-012 asks about 1,800 mm)	1714 mm <= 1800 mm	Within known science	Open
B8	Option B: two people turn about 1.65 drums an hour into flat panels	Mathematics	Orders of magnitude	15.1 m of cut at a 40 mm bite and 5 s a stroke (377 strokes), plus purge 14, loading 2, handling 6, three stretches of 4 min, ends 3 and deburring 4.4 min	claimed 1.65 drums/h, estimate 1.65 drums/h (ratio 1.00)	Within known science	Open

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**. A plausible row is closed when DMP-CAL-001 gives its number on paper; the coefficient it rests on is measured at TRL 4. Rows S4, S10, S16, S19 and S25 failed as first written and are closed by fixes F1 to F5.

Two notes on what the table does not say:

- S4 rests on the acceptance rule (only drums labelled for non-volatile oils with flash points above 60 °C are taken in) and on the reading at both bungs. The 60 °C limit is the upper bound of GHS flammable liquid category 3; liquids with flash points from 60 to 93 °C (GHS category 4) are still accepted by that rule. The vapour check catches any vapour they leave, so the claim holds; whether to tighten the rule to 93 °C is open decision 2 for Amish in DMP-DEC-001, not changed here.
- S11 was worded in DMP-CAL-001 v0.2 as strain "far past what the sheet can take without tearing or buckling". Low-carbon drum steel stretches well past 8.5 % before it tears; what the slip roll cannot do is put that strain in evenly, so a rolled hoop leaves the panel buckled. The design is unchanged (the hoops are cut out); the wording is corrected in DMP-CAL-001 v0.3, and the row is a design rationale.

Fields not relevant

- Physics, electromagnetism and circuits, optics and radiation, nuclear and ionising radiation: the bench set has no electrical, optical or radioactive parts; the gas detector is a bought instrument used to its manual.

- Chemistry, electrochemistry and batteries: the only battery is in the bought gas detector.
- Biology, microbiology, ecology and agronomy, plant and animal biology, epidemiology: no living material is processed; purge water is settled, skimmed and reused (S19).
- Engineering and technology, power and link budgets, control, sensing resolution: no power or control system; the detector's resolution is the bought instrument's (its alarm is set at 5 % LEL).

Concept options

- **Option A, bench set: rotary shears and slip roll** (the current design; recommended, proposed, awaiting Amish): rows S1 to S28, within known science or plausible and closed by DMP-CAL-001, with fixes F1 to F5. Pass.
- **Option B, lever shear and stretch leveller**: rows B1 to B8, within known science at first order. A 1,050 mm lever through a link gives about 22 to 1 at the cut, so one person shears 1.5 mm steel at about 122 N (B1); a 15 t hand-pumped cylinder stretches the widest, thickest panel 2 % past yield at 92.7 kN (B2), which wipes out the curl (B3); the wedge grips are set at 7 degrees so the serrated face out-grips the greased back (B6, corrected after the review). Its costs are elsewhere: each panel loses about 86 mm of length to the grips (B7, short of the 1,800 mm of PR-012) and the many short lever strokes make it slower, about 1.65 drums an hour (B8). Pass at first order; a calculation note would be written only if Amish chose it.

Independent review

On 2026-10-24 an independent reviewer (a second agent, run headless, that saw only this check, DMP-CAL-001 v0.2, the product requirements and `docs/04-calcs/sizing.py`, not the author's reasoning) tried to refute the headline results and every "Within known science" verdict. It could not run code and recomputed by hand; every number it recomputed from the stated inputs reproduced. Its findings and what changed:

Table 2. Independent review findings.

Row or result	Finding	Severity	Change
S10, R5 flatness	A reading found once per batch does not hold: 0.05 mm more wall gives about 20 mm of sag over 1 m, 10 MPa more yield about 16 mm	Refutes	Rechecked in the script (0.02 mm or 5 MPa gives about 8 mm). Fix F1: straightedge check of the first panel of every drum, dial trim, second pass when out of tolerance; 1.5 minutes a drum; PR-005, CAL section 5.3
Throughput	One drain stand is held for the whole purge cycle (about 28 to 37 minutes), capping the set below the 2.26 drums an hour claimed; the slit and ring times divide the disc speed by 3 without saying why; the elapsed purge was hard-coded	Weakens	Confirmed: 35.3 minutes a cycle, 1.7 drums an hour for one stand. Fix F2: a second stand and tray (S25 added). The factor of 3 is now documented as an effective speed allowance; the hard-coded fill removed. Rate 2.20 drums an hour with F1
S4	A catalytic sensor calibrated on methane under-reads heavy hydrocarbon vapour (relative response about 0.3 to 0.5)	Weakens	Fix F3: LEL channel calibrated on pentane or set with the maker's correction factor; BOM line 4, PR-002
S16, guards	8.0 mm slots sit on the 8 mm boundary; at 8.1 mm the distance needed is 80 mm	Weakens	Fix F4: slots toleranced 7 to 8 mm as made; PR-010
S19	Fire-resistant hydraulic fluids sink in or mix with water; detergent slows settling	Weakens	Fix F5: those fluids refused at acceptance (PR-013); the water settles overnight
S6, S13	Half the 1.5 mm feed resistance is an unmeasured edge-spreading estimate; doubled, two ring cuts need about 208 N and the disc slips; shear strength is not scaled with yield	Weakens	Rows stay plausible; the spreading force added to the items to confirm (DMP-DEC-001, Table 2, item 10); the hand push on the trolley is the fallback
S7	The raked-edge estimate assumes edges longer than the 42.5 mm a 7.5 mm stack needs; the side edges are 35 mm; seam work hardening ignored	Weakens	Row stays plausible; noted on the scrap-chime test (DMP-DEC-001, Table 2, item 8)
S9	The bending roll height assumes a circular arc; a linear moment gives about 9.7 mm nominal (6 to 13 mm)	Weakens	Noted in CAL section 5.2; the trial pass finds the setting; the travel claim stands
S14	The 25 kg reference mass holds only for ideal lifting; lifting the 23.7 kg head to the rail is above a realistic limit	Weakens	PR-011 and the row: both heads are two-person lifts to the rail (as the build plan already had it)
S17	The 2 mm heated strip is not a bound; a narrower shear zone gives a higher local rise	Weakens	Row reworded and made plausible; the vapour reading is the safeguard; edge temperature to measure at TRL 4
S1	The README's three sheets of 460 x 1,830 mm per drum (2.5 m2) cannot come from one drum (body 1.58 m2); the check compared the claim with itself	Weakens	README corrected; the row now checks against the body area
S3	The first-order check used a wrong mean head	Weakens (method)	The check now integrates the falling level (1.8 minutes to the bung centre against the CAL's 2.3 with the tail)

Row or result	Finding	Severity	Change
S11, S21, S24	A heuristic, a qualitative claim and an arithmetic check	Weakens	Reworded as a design rationale, a mechanism and an arithmetic check
S5, S12	Friction 0.3 on an oily chime and 0.15 on wet panels are screening values	Weakens	The end cutter clamp at friction 0.15 (about 1,040 N) added to the items to confirm; panels are wiped dry before rolling
B6	Self-locking is the wrong criterion; grip needs face friction above tan(angle) plus back friction	Refutes (as written)	Criterion corrected; wedge angle 7 degrees in the model
B4, B8	No pump efficiency; 1.65, not 1.7	Weakens	Efficiency 0.8 included (110 N); 1.65 drums an hour
S2, S8, S18, S20, S22, S23, B2, B3, B5, B7	Upheld	Upheld	None (B3: Luders bands and the 1 % loss of width noted for option B)
Missing claims	Rail beam deflection, roll deflection, punch jaw stress, purge stand capacity, soak adequacy, purge water reuse, head disc flatness, the punch die zone, option B stored hydraulic energy and grip release		Rows S25 to S28 added; soak adequacy is what the vapour check decides (a second purge on a failed check); the punch die zone and option B's stored energy are in DMP-RSK-001

The reviewer also found that the copy of DMP-CAL-001 it saw (v0.2) still had the old S11 wording and named `components()`, and that the product requirements listed two revisions numbered 0.6: all three are corrected (DMP-CAL-001 v0.3, DMP-PRS-001 v0.6).

Sources

- 29 CFR 1926.352(i), Fire prevention (drums, containers or hollow structures that held toxic or flammable substances are filled with water or thoroughly cleaned, ventilated and tested before welding, cutting or heating), Legal Information Institute, https://www.law.cornell.edu/cfr/text/29/1926.352, accessed 2026-10-25.
- 29 CFR 1910.252(a)(3), Welding, cutting and brazing: used containers and venting and purging, Legal Information Institute, https://www.law.cornell.edu/cfr/text/29/1910.252, accessed 2026-10-25.
- 29 CFR 1915.14(b), Hot work (no hot work at or above 10 percent of the lower explosive limit), Legal Information Institute, https://www.law.cornell.edu/cfr/text/29/1915.14, accessed 2026-10-25.
- American Welding Society, *F4.1:2017 Safe Practices for the Preparation of Containers and Piping for Welding, Cutting, and Allied Processes* (cleaning by water, steam, chemicals and inerting), https://pubs.aws.org/p/1759/f412017-safe-practices-for-the-preparation-of-containers-and-piping-for-welding-cutting-and-allied-processes, accessed 2026-10-25.
- Workplace Health and Safety Queensland, "Flammable and Combustible Liquids" (GHS flammable liquid categories: category 3 flash point 23 to 60 °C, category 4 above 60 to 93 °C), https://worksafe.qld.gov.au/safety-and-prevention/hazards/hazardous-chemicals/specific-hazardous-chemicals/flammable-and-combustible-liquids, accessed 2026-10-25.
- ISO 13857:2019, Safety of machinery: safety distances to prevent hazard zones being reached by upper and lower limbs, Table 4 (slot openings), https://www.iso.org/standard/69569.html, accessed 2026-10-25.

- ISO 11228-1, Ergonomics: manual handling, lifting and carrying (reference mass 25 kg), as cited for the focus area in CRP-REG-001 (checked 2026-10-24); EN 1005-3, recommended force limits for machinery operation (assessment at TRL 4).
- Trick-Tools, "Beverly B-1 Throatless Shear, 14 gauge capacity" (a lever throatless shear rated for 14 gauge, about 1.9 mm, mild steel), https://www.trick-tools.com/Beverly_B_1_Throatless_Shear_14_gauge_capacity_B_1_1116, accessed 2026-10-25.
- DMP-CAL-001 v0.3 (`docs/04-calcs/01-sizing.md`, `docs/04-calcs/sizing.py`) for every option A number; textbook relations: rotary and inclined-blade shear, elastic and perfectly plastic bending and springback, Torricelli outflow, wedge self-locking, adiabatic heating.